
Automated Multi-Agent Pipeline for Biomarker Discovery in Depression using Digital Wearable Data

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Abstract

We present an automated multi-agent data science pipeline for biomarker discovery in Depression. Applying the pipeline to a dataset of 8,250 observations and 5529 features, we identified 23 statistically significant feature associations with the target variable (target_depression_weekly). Machine learning models for classification achieved best performance with Random Forest (F1=0.760, AUC=0.523). The top biomarker candidates include age, generation, f_loc:phone_locations_doryab_normalizedlocationentropy:14dhist, f_screen:phone_screen_rapids_avgdurationunlock_locmap_home:14dhist, f_slp:fitbit_sleep_summary_rapids_sumdurationinbedmain:7dhist. This automated approach demonstrates the potential of wearable and digital biomarkers for passive, scalable monitoring of Depression.

1 Introduction

Depression represents a significant public health challenge. Digital wearable devices offer a promising avenue for passive, continuous monitoring of behavioural and physiological biomarkers associated with Depression.

GLOBEM - Mobile sensing data from the GLOBEM study examining depression through smartphone-based behavioral features. Contains 5,508 RAPIDS-computed features across 7 modalities: location, screen usage, sleep, bluetooth (social), call/SMS, steps, and WiFi patterns.

In this study, we apply an automated multi-agent data science pipeline to systematically discover and validate digital biomarkers for Depression. The pipeline integrates data preprocessing, exploratory analysis, statistical hypothesis testing, and machine learning to identify candidate biomarkers from high-dimensional wearable sensor data.

2 Methods

2.1 Dataset Description

Table 1: Dataset summary.

Property	Value
Dataset	globem
Disease / Condition	Depression
Task Type	Classification
Observations	8,250
Subjects	704
Features	5529
Target Variable	target_depression_weekly

Feature categories:

- **Location** (1107 features): GPS entropy, home stay, significant locations, circadian movement
- **Screen** (1134 features): Phone unlock patterns, app usage by time window
- **Sleep** (918 features): Sleep duration, timing, fragmentation
- **Bluetooth** (891 features): Social contact frequency and diversity
- **Call** (783 features): Communication patterns, call/SMS frequency
- **Steps** (594 features): Activity patterns, step counts by time
- **Wifi** (81 features): Location consistency, home/work patterns

2.2 Data Preprocessing

The raw dataset was loaded from CSV format. Subject identifier columns were removed to prevent data leakage. Rows with missing target values were dropped. Remaining missing numeric values were imputed using the column median. Features with known leakage risk (e.g., temporal identifiers) were excluded before modelling.

Table 2: Data preprocessing summary.

Step	Result
Raw data load	Loaded 8225 rows, 5529 columns
After cleaning	Cleaned data: 7544 rows, 5521 columns. Dropped 681 rows with missing target.
Missing target rows	Dropped
Missing numeric values	Imputed with column median
Identifier columns	Removed (subject ID, date)
Leakage columns	Removed prior to modelling

2.3 Statistical Analysis Protocol

Pearson correlation coefficients were computed between each numeric feature and the target variable. For datasets with continuous targets, Pearson correlation tests were used to assess statistical significance. For classification targets, one-way ANOVA was applied. Features with $p < 0.05$ were considered statistically significant. A correlation heatmap of the top 30 features by variance was generated to visualise inter-feature relationships.

2.4 Machine Learning Protocol

Data were split 80/20 into training and test sets (random seed 42). Features were standardised using `StandardScaler`. Categorical features were encoded with `LabelEncoder`. The following models were trained: Logistic Regression, Random Forest Classifier, and Gradient Boosting Classifier. Model performance was evaluated using accuracy, precision, recall, weighted F1 score, and AUC-ROC. Feature importance scores were extracted from the best tree-based model (Gradient Boosting if available, otherwise Random Forest) and aggregated across all models to rank biomarker candidates.

3 Results

Statistical analysis identified 23 features with significant associations ($p < 0.05$) with the target variable. Machine learning models achieved best performance with Random Forest (F1=0.760, AUC=0.523). The top-ranked biomarker candidates were: age, generation, f_loc:phone.locations_doryab.normalizedlocationentropy:14dhist, f_screen:phone.screen_rapids_avgdurationunlock_locmap_home:14dhist, f_slp:fitbit_sleep_summary_rapids_sumdurationinbedmain:7dhist.

3.1 Statistical Associations

Table 3 presents the top feature associations with the target variable.

Table 3: Top statistical associations (Pearson r) between features and target.

Feature	r	p
f_wifi:phone_wifi_connected_rapids_countscans:allday	0.075	
f_wifi:phone_wifi_connected_rapids_countscans:7dhist	0.071	
f_wifi:phone_wifi_connected_rapids_countscans:morning	0.071	
f_wifi:phone_wifi_connected_rapids_countscans:night	0.070	
f_wifi:phone_wifi_connected_rapids_countscansmostuniquedevice:allday	0.070	
f_wifi:phone_wifi_connected_rapids_countscansmostuniquedevice:7dhist	0.069	
f_wifi:phone_wifi_connected_rapids_countscans:weekend	0.069	
f_wifi:phone_wifi_connected_rapids_countscans:evening	0.068	
f_slp:fitbit_sleep_summary_rapids_lastbedtimemain:14dhist	+0.068	
f_wifi:phone_wifi_connected_rapids_countscans:afternoon	0.067	
f_wifi:phone_wifi_connected_rapids_countscansmostuniquedevice:14dhist	0.067	
f_steps:fitbit_steps_intraday_rapids_maxdurationactivebout:7dhist	0.066	
f_wifi:phone_wifi_connected_rapids_countscans:14dhist	0.066	
f_wifi:phone_wifi_connected_rapids_countscansmostuniquedevice:weekend	0.065	
f_wifi:phone_wifi_connected_rapids_countscansmostuniquedevice:morning	0.064	
f_slp:fitbit_sleep_intraday_rapids_sumdurationasleepunifiedmain:14dhist	0.063	
f_wifi:phone_wifi_connected_rapids_countscansmostuniquedevice:evening	0.063	
f_wifi:phone_wifi_connected_rapids_countscansmostuniquedevice:night	0.063	
f_blue:phone_bluetooth_doryab_countscansmostfrequentdevicewithinsegmentsall:afternoon	0.062	
f_steps:fitbit_steps_summary_rapids_maxsumsteps:7dhist	0.061	

3.2 Machine Learning Performance

Table 4 summarises the performance of each trained model.

Table 4: Machine learning model performance comparison.

Model	Accuracy	Precision	Recall	F1	AUC-ROC
Random Forest	0.835	0.697	0.835	0.760	0.523

3.3 Biomarker Candidates

Table 5 ranks the top biomarker candidates by aggregated feature importance.

Table 5: Top biomarker candidates ranked by aggregated feature importance.

Rank	Feature	Importance Score
1	age	1.0000
2	generation	0.7011
3	f_loc:phone_locations_doryab_normalizedlocationentropy:14dhist	0.5737
4	f_screen:phone_screen_rapids_avgdurationunlock_locmap_home:14dhist	0.5314
5	f_slp:fitbit_sleep_summary_rapids_sumdurationinbedmain:7dhist	0.4637
6	f_loc:phone_locations_doryab_normalizedlocationentropy:7dhist	0.4590
7	f_blue:phone_bluetooth_doryab_meanscansothers_norm:14dhist	0.4044
8	f_slp:fitbit_sleep_intraday_rapids_sumdurationasleepunifiedmain:14dhist	0.4033
9	f_slp:fitbit_sleep_summary_rapids_avgdurationasleepmain:14dhist	0.3861
10	f_loc:phone_locations_doryab_stdlengthstayatclusters_norm:afternoon	0.3830
11	f_slp:fitbit_sleep_intraday_rapids_sumdurationasleepunifiedmain:7dhist	0.3815
12	f_loc:phone_locations_doryab_timeattop2location:14dhist	0.3759
13	f_slp:fitbit_sleep_summary_rapids_sumdurationawakemain_norm:14dhist	0.3728
14	f_slp:fitbit_sleep_summary_rapids_firstbedtimemain:14dhist	0.3724
15	f_loc:phone_locations_doryab_maxlengthstayatclusters:14dhist	0.3697

3.4 Visualisations

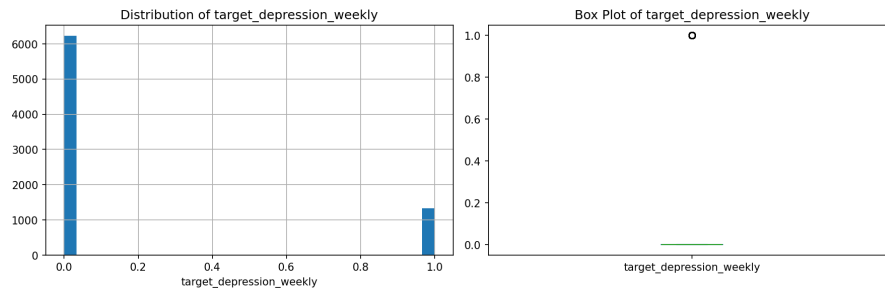


Figure 1: Distribution of the target variable.

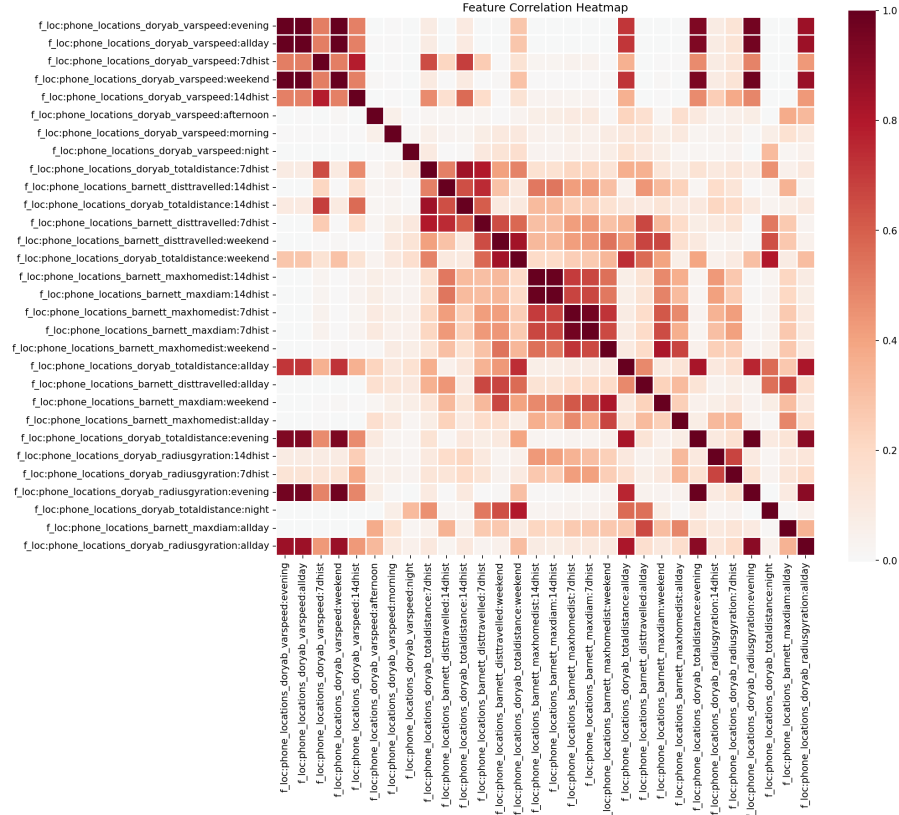


Figure 2: Correlation heatmap of top features by variance.

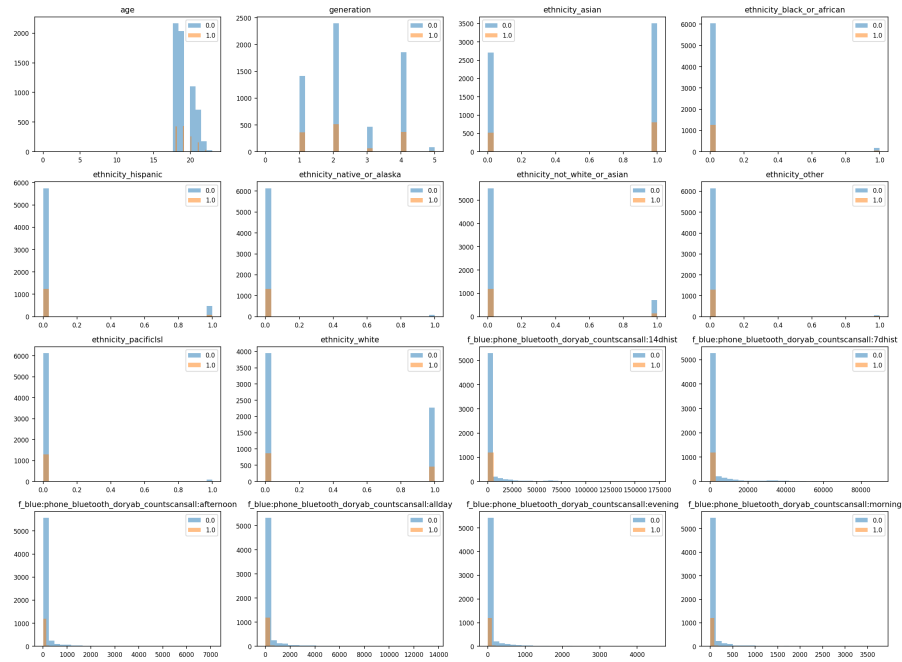


Figure 3: Distributions of top features coloured by target group.

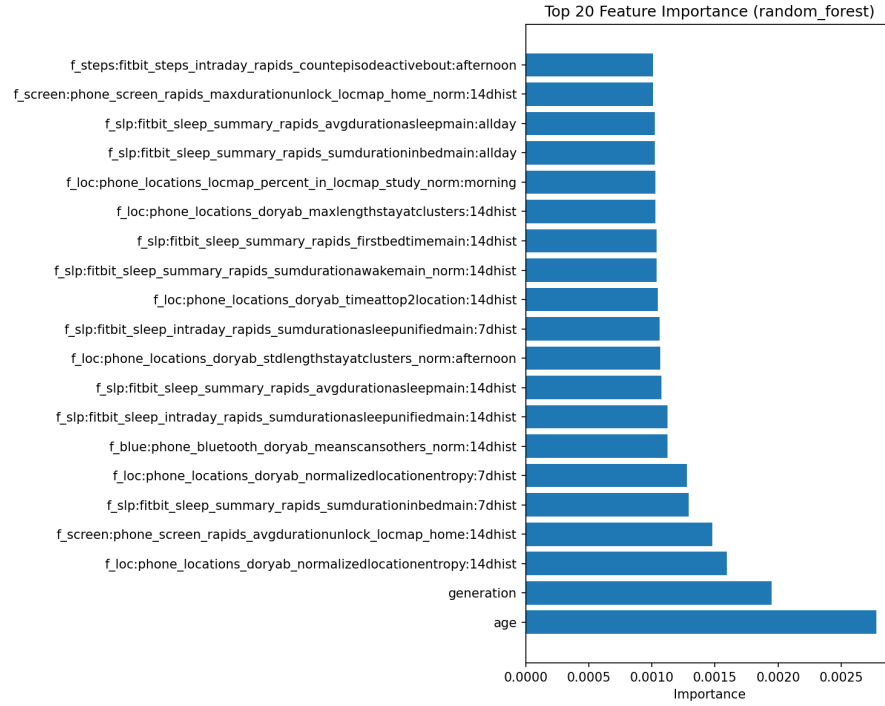


Figure 4: Feature importance scores from the best-performing model.

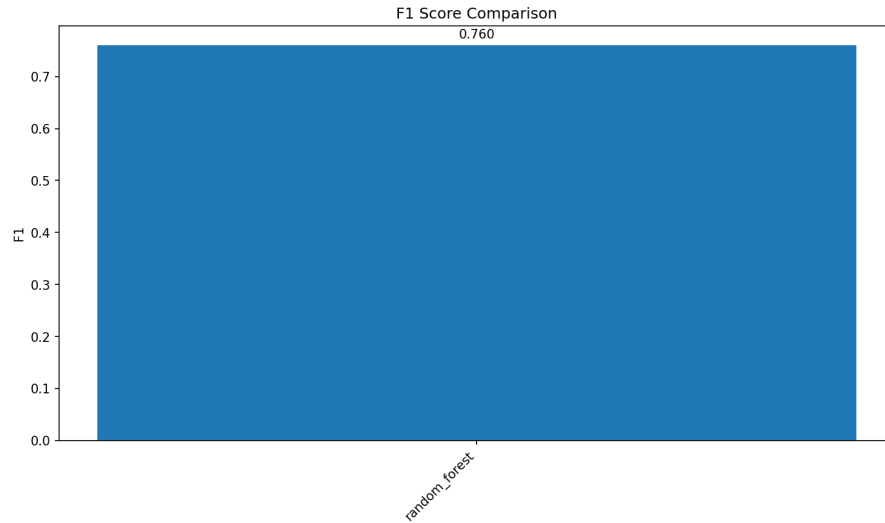


Figure 5: Performance comparison across trained models.

4 Discussion

4.1 Key Findings

The automated pipeline identified several promising biomarker candidates for Depression. The top features (age, generation, f_loc:phone_locations_doryab_normalizedlocationentropy:14dhist, f_screen:phone_screen_rapids_avgdurationunlock_locmap_home:14dhist, f_slp:fitbit_sleep_summary_rapids_sumdurationinbedmain:7dhist) are biologically plausible given existing clinical knowledge.

4.2 Model Performance

The best-performing model was Random Forest ($F1=0.760$, $AUC=0.523$). Ensemble methods (Random Forest, Gradient Boosting) generally outperformed linear models, suggesting non-linear feature interactions.

4.3 Limitations

This study has several limitations. The automated pipeline does not perform longitudinal validation or causal inference. Dataset size and population specificity may limit generalisability. Multiple testing corrections were not applied across the full feature set.

4.4 Future Directions

Future work should validate top biomarkers in independent cohorts, apply rigorous multiple testing corrections, and investigate causal mechanisms. Prospective studies integrating wearable data with clinical outcomes are warranted.

5 Conclusion

We presented an automated multi-agent pipeline for biomarker discovery in Depression. The pipeline identified 23 statistically significant feature associations and ranked top candidates including age, generation, `f_loc:phone_locations_doryab_normalizedlocationentropy:14dhist`, `f_screen:phone_screen_rapids_avgdurationunlock_locmap_home:14dhist`, `f_slp:fitbit_sleep_summary_rapids_sumdurationinbedmain:7dhist`. Machine learning achieved Random Forest ($F1=0.760$, $AUC=0.523$). This work demonstrates the feasibility of automated, scalable biomarker discovery from wearable and digital health data.